Multi-session Online Open-vocabulary Object Mapping

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Abstract

In this paper, we present a multi-session online open-vocabulary object mapping system. Robots operating in previously unseen, complex real-world environments can benefit from semantic scene representations that can be constructed online and queried using natural language, enabling them to reason about tasks in the context of their surroundings. Multi-session mapping capabilities allow robots to reuse, refine and extend maps over repeated deployments while also providing the foundation for collaborative mapping by a team of robots. However, existing open-vocabulary mapping systems typically either rely on precomputed camera poses and offline processing pipelines or have focussed on single-session operation, restricting their applicability to long-term autonomy. Our proposed system addresses this gap by integrating a robust sparse visual SLAM backend with an open-vocabulary object-mapping frontend. The resulting framework enables the incremental construction, refinement, and extension of open-vocabulary object maps over repeated deployments while maintaining consistency with the evolving SLAM estimate. We quantitatively evaluate geometric accuracy, object association, cross-session object consolidation and runtime performance on Replica, and demonstrate persistent multi-session mapping on a real-world RGB-D sequence.

Keywords: SLAM, Multi-session, Open-vocabulary

1 Introduction

For robots to operate effectively in unknown real-world environments, they must be able to reason about their surroundings at a semantic level. This requires the ability to construct scene representations that capture not only the geometry of the environment, but also information describing the objects it contains and their spatial relationships. Robot perception systems have evolved over time from producing geometry-centric maps designed primarily for localisation and reconstruction toward increasingly semantic models that support higher-level reasoning and interaction. Recent approaches leverage vision foundation models to produce open-vocabulary scene representations that are not restricted to a predefined set of object categories, allowing natural language queries or instructions to be grounded directly in the map (Gu et al., 2024). This makes such representations particularly attractive for robots that must operate in previously unseen environments, where the relevant objects and task descriptions cannot be fully specified in advance.

For practical robot deployment, it is critical that such scene representations are built incrementally and online. This capability allows a robot to build a map as it explores an environment for the first time, rather than relying on a pre-existing reconstruction or a sequence of accurately posed images. It also allows the representation to be refined when improved pose estimates become available, for example after loop closure and backend optimisation. Recent systems (Gassol Puigjaner et al., 2026; Gorlo et al., 2026) address online open-vocabulary scene representation by extending closed-vocabulary online hierarchical 3D scene graph systems such as Hydra (Hughes et al., 2022). However, these systems do not explicitly address multi-session mapping. Multi-session operation is a key capability for long-term autonomy, allowing a robot to reuse, extend, and refine its map over repeated deployments in overlapping regions of the same environment. It also provides a foundation for collaborative mapping with teams of robots, where observations collected by different agents or during different sessions can be integrated into a shared persistent representation.

In this paper, we address this gap by integrating the object-mapping component of ConceptGraphs (Gu et al., 2024) with ORB-SLAM3 (Campos et al., 2021), a mature sparse visual SLAM system that supports multi-session mapping. The resulting system incrementally constructs an online, multi-session, open-vocabulary object map from RGB-D image sequences. For each input sequence, the system estimates camera poses, detects open-vocabulary object observations, associates them across views, and fuses them into a persistent 3D object map. By coupling the object map to the SLAM backend, the system can update object positions and geometry when optimised camera poses become available, and can extend the object map across multiple mapping sessions. This enables persistent language-grounded object mapping in environments that are explored incrementally over time. The code will be available at `https://github.com/robotvisionmu/orb-cg`.

2 Related Work

Scene representations in robotics have progressively shifted from geometry-centric maps toward increasingly semantic and language-aware models of the environment. Sparse visual SLAM systems (Campos et al., 2021) represent the world using visually salient landmarks, providing an efficient structure for camera tracking and localisation. Dense mapping systems (Whelan et al., 2016) extend this representation by reconstructing detailed surface geometry of the environment. Semantic mapping (McCormac et al., 2018) further augments geometric maps with object labels, allowing robots to reason not only about where surfaces are, but also about what they correspond to.

3D scene graphs provide a more structured representation of indoor environments by combining metric, semantic, and topological information in a unified hierarchy. Armeni et al. (2019) proposed a seminal approach that uses a 3D mesh and registered panoramic images to construct a building-scale scene graph containing object, room, and camera semantics, as well as relationships between these entities. However, the representation is constructed offline from a precomputed reconstruction. Rosinol et al. (2021) extended this work by integrating visual-inertial SLAM to construct a metric-semantic mesh of the scene online. The system also tracks dynamic elements in the scene, such as humans and robots. The metric-semantic mesh is built online, but the higher-level layers of the dynamic scene graph are computed offline as a post-processing step after mapping. Hydra (Hughes et al., 2022) subsequently addressed this limitation by constructing the entire hierarchical structure online in real-time but remains closed-vocabulary.

Other works have explored the interaction between scene graph representations and SLAM backends. In particular, vS-Graphs (Tourani et al., 2026) integrates a scene graph representation with ORB-SLAM3, using semantics of structural scene elements (ground surfaces, walls, rooms) as constraints to improve SLAM performance. This demonstrates the potential benefits of coupling higher-level scene representations with visual SLAM. However, the output representation is limited in semantic scope to serve the primary goal of improving localisation and mapping performance through the integration of semantic constraints.

Recent work has extended scene representations toward open-vocabulary and language-grounded mapping. HOV-SG (Werby et al., 2024) constructs hierarchical open-vocabulary scene graphs that support language-based queries over objects and regions. ConceptGraphs (Gu et al., 2024) constructs object-centric 3D scene graphs from RGB-D observations by associating object detections across views and fusing them into a global 3D map. These systems demonstrate the value of open-vocabulary representations for querying and reasoning about 3D environments beyond a fixed semantic label set. However, they assume that accurate camera poses are already available and operate primarily offline or in a batch-processing setting. As a result, they do not directly address the problem of maintaining an open-vocabulary object map while a robot is actively exploring an unknown environment.

Recent systems also address the online open-vocabulary mapping problem. ReasoningGraph (Gassol Puigjaner et al., 2026) and Describe Anything Anywhere at Any Moment (DAAAM) (Gorlo et al., 2026) extend online 3D scene graph construction toward open-vocabulary mapping by building on the Hydra framework mentioned previously. Similar to our work, OVO (Martins et al., 2025) integrates an open-vocabulary mapping system with a SLAM backbone to maintain an open-vocabulary map of tracked 3D segments. However, these systems do not explicitly address multi-session mapping. In contrast, our work focuses on coupling open-vocabulary object mapping with a visual SLAM backend that supports multi-session operation, enabling a robot to explore an environment over multiple deployments and construct a globally consistent object map.

3 Methodology

The proposed system consists of a ConceptGraphs-based (Gu et al., 2024) open-vocabulary object-mapping frontend and ORB-SLAM3 (Campos et al., 2021) as a visual SLAM backend. An overview of the system architecture is shown in Figure 1. The system maintains a separate object-centric, multi-map scene representation that mirrors the map and keyframe structure of the ORB-SLAM3 *Atlas*. This Atlas-aligned representation is denoted by $A = \{M_i\}_{i=1}^{N}$, where each map $M_i = (K_i, O_i)$ corresponds to an ORB-SLAM3 map and contains a set of keyframes K_i and a corresponding set of mapped objects O_i. Each keyframe $k = (\mathbf{T}_k, \mathbf{I}_k, \mathbf{D}_k, Z_k)$ comprises its pose, RGB image, depth image, and a set of object observations Z_k. A mapped object $o = (P_o, \mathbf{f}_o)$ is formed by fusing a set of supporting observations Z_o, where P_o is its 3D point-cloud representation and $\mathbf{f}_o$ is its aggregated CLIP feature.

Each input RGB-D frame is first processed by ORB-SLAM3, which selects a subset of frames as keyframes to provide spatial coverage of the environment for efficient mapping and localisation. Rather than processing every input frame, the frontend processes only keyframes, which form the interface between the SLAM backend and the object-mapping mapping frontend. As ORB-SLAM3 performs loop closure, relocalisation, map merging, and pose optimisation, keyframe poses may be updated and keyframes may be reassigned to another map or removed. These changes are propagated to the frontend through keyframe events, allowing the corresponding object observations and mapped objects to remain consistent with the current SLAM estimate.

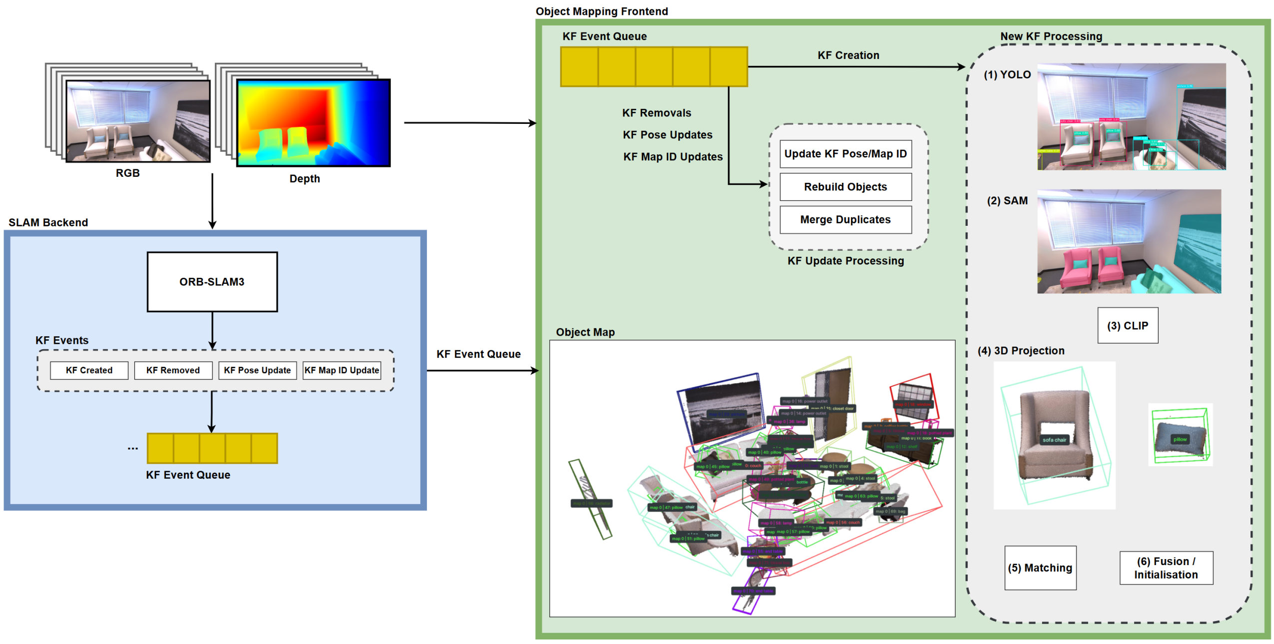


Figure 1: System architecture of the proposed ORB-SLAM3 and ConceptGraphs-based object-mapping pipeline. RGB-D frames are processed by ORB-SLAM3, which provides keyframe selection, camera tracking, pose optimisation, relocalisation, loop closure, and multi-session map management. Selected keyframes form the interface to the object-mapping frontend. A keyframe event queue propagates asynchronous SLAM state changes, including keyframe creation, removal, pose updates, and map reassignment. The frontend processes keyframes using open-vocabulary detection, segmentation, vision-language feature extraction, 3D projection, and object association to maintain a persistent open-vocabulary object map consistent with the current SLAM estimate.

3.1 SLAM Backend (ORB-SLAM3)

For each incoming RGB-D frame, ORB-SLAM3 estimates the camera pose relative to the active map and selects keyframes for mapping. Keyframe poses and map points are refined through local bundle adjustment, while place recognition enables loop closure, relocalisation, and map merging.

Of particular relevance to this work is the ORB-SLAM3 *Atlas*, which maintains multiple maps and a shared place-recognition database. An *Atlas* from a previous session can be loaded during initialisation, while a new map can be created when tracking cannot be recovered in an existing map. ORB-SLAM3 continues to search for correspondences with previously stored maps and merges them into a common coordinate frame when sufficient overlap is detected. This mechanism supports incremental multi-session operation, since maps constructed at different times, or from partially disconnected trajectories, can be reused and merged into a single persistent spatial representation of the environment.

3.2 Keyframe Event Interface

Since ORB-SLAM3 is a real-time, multithreaded system, its state can change asynchronously relative to the object-mapping frontend. To propagate these changes, we introduce a keyframe event queue. Events are emitted when a keyframe k is created or removed, when its pose $\mathbf{T}_k$ is updated, or when it is reassigned to another map following relocalisation or map merging. These events allow the frontend to update the corresponding observations Z_k and their map associations. While an event is waiting to be consumed, subsequent events for the same keyframe are coalesced, avoiding unnecessary reprocessing while ensuring that the frontend receives the most recent SLAM state.

3.3 Open-Vocabulary Object Mapping

The frontend is based on the open-vocabulary object-mapping component of ConceptGraphs. For each keyframe k, its RGB-D images, camera intrinsics, and estimated pose $\mathbf{T}_k$ are retrieved. YOLO-World (Cheng et al., 2024) extracts open-vocabulary object detections from the RGB image, and SAM (Kirillov et al., 2023) generates corresponding object masks. Detections are filtered to remove low-confidence, very small, background, or otherwise invalid regions. CLIP (Radford et al., 2021) vision-language features are computed for the remaining object proposals.

The resulting detections form the object observations Z_k. Each valid observation is projected into 3D using the depth image and camera intrinsics and transformed into the coordinate frame of its associated map using $\mathbf{T}_k$. The observation is then compared with existing objects $o \in O_i$ using spatial overlap and visual-semantic similarity. If a suitable match is found, the observation is added to the object's supporting observations Z_o and fused into its geometry P_o and CLIP feature $\mathbf{f}_o$. Otherwise, a new mapped object is created.

Unlike the full ConceptGraphs pipeline, the proposed system maintains an object-level semantic representation rather than constructing a complete relationship graph. Relationship extraction is omitted because it is task-dependent and unnecessary for maintaining the persistent object map. The resulting representation therefore consists of open-vocabulary object instances with 3D geometry and aggregated vision-language features.

3.4 Updating the Object Map After SLAM Optimisation

When ORB-SLAM3 updates the pose $\mathbf{T}_k$ of a keyframe, the change is propagated to the frontend through a keyframe event. Any mapped object o supported by observations from the affected keyframe is marked for reconstruction. Its geometry P_o is then rebuilt from the supporting observations Z_o using the latest keyframe poses, ensuring consistency with the optimised SLAM estimate. If a keyframe is removed, its observations are removed from the corresponding supporting sets Z_o. Objects with no remaining valid observations are deleted. Duplicate objects within the same map are subsequently checked and merged where appropriate.

3.5 Multi-Session Mapping

The proposed system leverages the multi-map capabilities of the ORB-SLAM3 *Atlas*. When ORB-SLAM3 creates a new map, for example following tracking loss or at the start of a new session, the frontend maintains a corresponding map $M_j = (K_j, O_j)$ while preserving the object sets associated with previous maps. Object observations are therefore accumulated only within the map to which their originating keyframes currently belong.

If ORB-SLAM3 subsequently detects overlap and merges two maps, the resulting keyframe pose and map-assignment changes are propagated to the frontend through the event interface. The corresponding mapped objects are transferred into the resulting map and compared using the same spatial and visual-semantic association criteria applied to new observations. Duplicate representations of the same physical object can therefore be consolidated after the underlying SLAM maps have been aligned and merged.

This design allows object maps to be extended over repeated deployments in overlapping regions of the same environment. It also provides a basis for collaborative mapping, where observations collected during different sessions, or potentially by different robots, can be integrated into a shared persistent representation once their underlying SLAM maps are aligned or merged.

4 Evaluation

The proposed system is evaluated in both single-session and multi-session settings. Single-session evaluation measures the geometric accuracy of the constructed object map and the consistency of object association, using seven RGB-D sequences of the 3D Replica dataset (Straub et al., 2019) provided by iMAP (Sucar et al., 2021), for which ground-truth camera poses and semantic instance annotations are available. The `room 2` sequence was excluded because tracking loss caused ORB-SLAM3 to initialise a second map, preventing a single rigid trajectory alignment. Multi-session evaluation measures the system's ability to reconcile and extend object maps across sessions. Finally, runtime performance isevaluated on the Replica sequences. The evaluation focuses the performance of the integration of ORB-SLAM3 and the ConceptGraphs-based frontend, not the individual performance of each system.

4.1 Single-Session

To prevent geometric accuracy metrics from being confounded by errors in object fusion, geometric performance is evaluated separately from object association. Geometric evaluation measures whether the object observations correctly represent the position and spatial extent of the corresponding physical object. Object association evaluation measures whether detections from the same physical object are grouped together without merging distinct objects or fragmenting one object across multiple map objects. For each Replica scene, visible ground-truth (GT) object geometry was reconstructed using semantic-instance images, depth images and GT camera poses corresponding to the keyframes selected by the system. Structural scene elements (walls, floors and ceilings) were excluded as these were not mapped by the system. Each object observation was assigned to the GT instance occupying the largest fraction of its predicted mask, provided this fraction was at least 0.5. GT masks were used only to establish identity and not to construct the predicted geometry. The ORB-SLAM3 trajectory was aligned to the Replica frame using an SE(3) transformation fitted between corresponding camera centres. The mean camera-centre residual after alignment was 0.90cm, providing context for the contribution of trajectory error to the reported geometry metrics.

For geometric evaluation, observation point clouds were grouped by assigned GT instance and transformed into the Replica frame. Centroid error measures 3D localisation, while AABB IoU measures agreement in object position and spatial extent. Table 1 reports pooled results over the seven Replica sequences. Of the 372 evaluable GT instances, 262 (70.4%) were assigned to object observations. Median centroid error was 3.92cm, compared with a mean of 10.82cm. Removing the worst 5% of instances reduced the mean to 7.13cm, showing that a small number of large errors strongly affected the average. These errors were primarily associated with incomplete or oversized masks spanning neighbouring instances. Mean 3D AABB IoU was 0.499, indicating additional errors in reconstructed object extent and overlap.

Object association was evaluated from the GT assignments of each mapped object's supporting observations. Of the 258 mapped objects, 33 had no observations assigned to a GT instance. Of the remaining 225 mapped objects, 154 (68.4%) were strictly clean, with every supporting detection assigned to the same GT instance. A further 55 (24.4%) contained confirmed support from multiple GT instances, while 16 (7.1%) combined single-GT support with detections for which no GT identity could be established. Additionally, 48 of the 262 (18.3%) GT instances were assigned to observations distributed across multiple mapped objects. The association results therefore show both over-merging and fragmentation, despite most mapped objects comprising observations of a single physical instance.

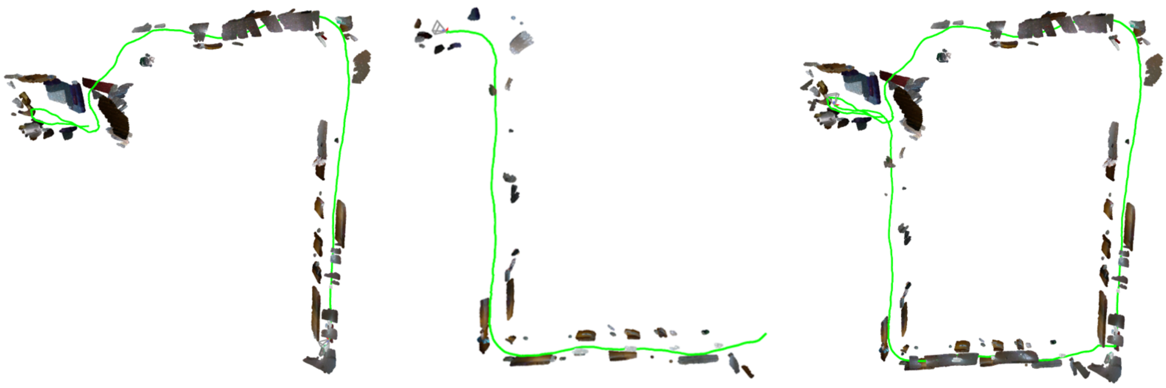
Table 1: Pooled single-session evaluation results on Replica

Geometry		*Object Association*	
Evaluable GT Instances	372	Mapped Objects	258
Assigned GT Instances	262	Unassigned	33
Alignment Residual (mean)	0.9cm	Clean	154 / 225 (68.4%)
Centroid (mean | median)	10.82cm | 3.92cm	Mixed	55 / 225 (24.4%)
AABB IoU (mean | median)	0.499 | 0.486	Partially Assigned	16 / 225 (7.1%)
		Fragmented GT	48 / 262 (18.3%)

4.2 Multi-Session

Cross-session object association was evaluated on the Replica `room 0` iMAP sequence, split into two sessions. The incoming-object association recall is reported, which measures how effectively objects from the second session are incorporated into the merged object map. Observations supporting each pre-merge mapped object were assigned to GT instances using the same procedure as in the single-session evaluation. A second-session object was eligible for consolidation if at least one of its observations shared a GT instance with an observation supporting a first-session object. Consolidation was successful when the second-session object and a corresponding first-session object were merged into the same final mapped object. Of the 22 eligible second-session objects, 19 were successfully consolidated (86.4%). As this metric operates on object predictions produced independently in each session, it cannot completely isolate cross-session consolidation from errors introduced during single-session mapping. Fragmentation or incomplete object reconstruction can alter both the number of eligible second-session objects and the evidence available for their subsequent association.

Multi-session persistence and map expansion were demonstrated using two RealSense RGB-D sessions captured along partially overlapping trajectories in the Eolas Building at Maynooth University. Figure 2 shows the first session map is retained and extended by the second session following the ORB-SLAM3 map merge. This demonstrates the construction of a persistent object map over repeated deployments.



(a) (b) (c)

Figure 2: Multi-session object mapping on real-world scene captured by RealSense. Two partial maps (a) and (b) are constructed from separate sessions with overlapping trajectories and are subsequently fused into a single persistent object map (c).

4.3 Runtime

Runtime was measured over the seven Replica sequences on an Intel Core i7-13700KF CPU, 32 GB RAM, and an NVIDIA RTX 4090 GPU. The sequences comprise 14,000 RGB-D frames at 1200×680 resolution, from which ORB-SLAM3 selected 1,294 keyframes. The timings are reported in Table 2. ORB-SLAM3 tracking, which is executed for every input frame, averaged 12.48 ms per frame. Synchronous object-mapping processing (YOLO-World, SAM, CLIP, 3D projection and object matching) averaged 493.4 ms per keyframe. The dominant costs were SAM segmentation (158.9 ms) and CLIP embedding (224.6 ms). Object-map maintenance, including geometry rebuilding and duplicate-object merging, required an additional 204.1 ms of computation when normalized by the number of processed keyframes. This value is an amortized computational cost rather than synchronous keyframe latency, since maintenance operations are triggered separately as the map is updated.

Table 2: Mean runtime on Replica

Component	*Time*
ORB-SLAM3 tracking	12.48 ms/frame
Synchronous KF processing	493.42 ms/KF
Map maintenance	204.12 ms/KF

5 Conclusion

In this paper, we presented a system for multi-session online open-vocabulary object mapping that integrates ORB-SLAM3 with a ConceptGraphs-based object-mapping frontend. By coupling object observations to SLAM keyframes, the system incrementally constructs an object map, updates its geometry as the SLAM estimate changes, and retains and reconciles object maps across mapping sessions.

The evaluation demonstrates that the system generally achieves accurate object localisation, although reconstruction of object spatial extent remains less consistent. Object association also remains a source of error, with both over-merging and fragmentation evident. These errors can propagate to cross-session consolidation, since the reconciliation process operates on the object predictions produced independently in each session. Nevertheless, the multi-session experiments demonstrate that previously mapped objects can be retained, corresponding objects consolidated, and the object map extended over repeated deployments. Runtime analysis shows that object mapping introduces a significiant computational cost relative to SLAM tracking.

Future work will initially address object reconstruction and association (spatial extent, fragmentation and overmerging). Further work will focus on a multithreaded architecture to reduce mapping overhead before extending system towards simultaneous collaborative mapping, environmental change detection, application-specific relationships and hierarchies, ROS integration and support for the monocular and stereo configurations provided by ORB-SLAM3.

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